

Varieties of headless relative clauses in Tigrinya

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Abstract: This paper presents the first systematic description of headless relative clauses in Tigrinya (Semitic; Eritrea and Ethiopia; SOV). Drawing on novel fieldwork data, it examines three types of headless relatives, namely light-headed, super-free, and free-choice relatives. The data show that, unlike many other languages, Tigrinya forms both headed and headless relative clauses using the same morphosyntactic strategy, which involves the relativizer *z(i)-* without overt *wh*-elements. Headless relatives in Tigrinya do not rely on interrogative *wh*-structures, in contrast to languages such as English. Notably, free relatives are not attested. Instead, all headless relative clauses share a structural core with headed relatives, differing only in the presence or absence of nominal and determiner layers. By documenting this pattern, the paper identifies a previously underdescribed strategy in Afro-Asiatic. These findings refine the typology of relativization, provide empirical evidence for cross-linguistic variation in relative clause formation, and establish a foundation for formal analyses of Tigrinya and related languages.

Keywords: Tigrinya, Semitic, relative clauses, headless relative clauses, light-headed relative clauses, *wh*-words

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1 Introduction

Both the descriptive and typological literature have mainly focused on relative clauses (henceforth, RCs) of one kind — *restrictive (externally) headed relative clauses*. An example from English is given in (1) with the nominal head underlined, the RC in brackets, and the underscore marking the missing constituent in the RC (the subject).

(1) I bought [DP *the* [NP stuff [RC that — is on the table]]].

The RC combines with its head *stuff* to form an NP, which is then embedded under the Determiner Phrase (DP) with the definite determiner *the* — in italics in (1) as the Determiner (D) head. The RC syntactically and semantically behaves like a restrictive adjective: it semantically restricts the meaning of the NP it syntactically combines with. For instance, in (1), the speaker states that they didn't buy stuff in general, but the more specific stuff that is on the table.

Headless relative clauses have received much less attention. Like headed RCs, headless RCs are full clauses with a missing constituent; unlike headed RCs, headless RCs lack a nominal head. Instead of conveying some form of factual/propositional content as most other clauses do, headless RCs exhibit the same syntactic distribution and semantic behavior as DPs (or PPs), rather than restrictive adjectives. Two examples of headless RCs in English are given in (2), each of which exhibits the same distribution and interpretation as the complex DP in (1) which contains the headed RC and its nominal head.

- (2) a. I bought [_{RC} **what** — is on the table].
 b. I bought [_{DP} *that* [_{RC} which is on the table]].

The bracketed clause in (2a) is known as a *free relative clause* (henceforth, FR). It is characterized by being introduced by a *wh*-word that occurs in interrogative clauses as well (see Caponigro 2003, 2004, 2021). The bracketed clause in (2b), instead, is known as a light-headed relative clause (henceforth, LHR). Unlike an FR and like a headed RC, an LHR is embedded under the demonstrative *that*, which occurs as a D without a nominal complement and, as such, acts as the ‘light head’ of the LHR. Like an FR and unlike a headed RC, an LHR lacks a nominal head.¹

A third kind of headless RC is called *super-free relative clause* (henceforth, SFR) (Caponigro 2021, Caponigro & Fălăuş 2023). It is not attested in English, but in many other languages, including Maltese (Semitic, SOV), as shown in (3).

- (3) għamil-t [li għid-t-l-i]
 do.PFV-SM.1S COMP say.PFV-SM.2S-DAT-OM.1S
 ‘I did what you told me.’ (adapted from Sadler & Camilleri 2018:125,(31))

SFRs lack the nominal head, like all headless RCs, but also a D element and/or a *wh*-word introducing them, which is what distinguishes them from FRs and LHRs.

Headed RCs have been studied and analyzed far more extensively than headless RCs. As a result, headless RCs remain comparatively under-investigated, despite the theoretical insights they offer into the syntax and semantics of relativization and nominal interpretation. However, recent work has begun to investigate case studies of headless RCs in specific languages (a.o. de Vries 2002, Citko 2004, Caponigro 2003, 2004, 2021, Caponigro et al. 2021), shedding light on their syntactic behavior, semantic interpretation, and cross-linguistic variation.

As a contribution to the growing body of work on headless RCs, this paper provides the first (to our knowledge) systematic description of headless RCs in an Ethio-Semitic language. In doing so, it also highlights a pattern that we have not seen described before for a language with both headed and headless RCs. Tigrinya makes use of the same morphosyntactic device to form both headed and headless RCs — a device that does not rely on (interrogative) *wh*-clauses nor *wh*-words. Tigrinya has headed RCs and headless RCs of the kinds LHRs and SFRs that morpho-syntactically resemble headed RCs rather than interrogative *wh*-clauses. No FRs are attested. This behavior contrasts sharply with languages like Romanian (Romance; Caponigro & Fălăuş 2023) or English (Germanic), which lack SFRs completely and make use of the *wh*-words in interrogative clauses to form non-interrogative *wh*-clauses. It also differs from languages like Maltese (Semitic), which exhibits both FRs and SFRs (Sadler & Camilleri 2018). Finally, Tigrinya differs from Adyghe and Abaza (Northwest Caucasian): like Tigrinya, Adyghe and Abaza form headed and headless RCs using the same morphosyntactic device, without using any *wh*-words. Therefore, no FRs are attested. On the other hand, neither Adyghe nor Abaza allows for embedded interrogative *wh*-clauses, but both languages use RCs to convey interrogative meaning (Caponigro & Polinsky 2011, Arkadiev & Caponigro 2021). Tigrinya, instead, makes use of a full set of *wh*-words and *wh*-clauses to form (embedded) interrogative clauses.

¹The example in (i) shows a case in which a PP, an FR, and an LHR exhibit similar distribution and interpretation.

- (i) Tell me why I can’t be [_{PP} **in the place** you are in] / [_{FR} **where** you are] / [_{LHR} **there** where you are].

The version of (i) with the LHR is from the lyrics of the 1999 song *Show Me The Meaning Of Being Lonely* by Backstreet Boys.

Unless otherwise noted, the data presented in this paper were collected by the first author through individual interviews with three native Tigrinya-speaking consultants from the Debub and Gash-Barka regions of Eritrea. These speakers are proficient in French and have been living in Geneva since the 2010s. While we observed some regional linguistic variation among our consultants, we focus here on phenomena that appear to be largely consistent across these speakers.

Transcription conventions in Tigrinya are subject to significant individual variation. We have tried to account for this fact by using consistent glossing practices following the Leipzig Glossing Rules. Our only two deviations are using capital SM and capital OM respectively, for subject- and object-marking morphology on verbal elements and hyphenating ϕ -features in front of verbs in the Imperfective. We follow the guidelines proposed by [Meyer & Treis 2021](#) for citing Eritrean and Ethiopian authors in a way that respects their naming conventions. Accordingly, authors from these two regions are cited by their first names, unless there is evidence that they cite their own work using their family name.

The article is structured as follows. Section 2 provides background on Tigrinya that is relevant to the main topic of the paper, namely the morphosyntax of nominal modifiers, interrogative elements and clauses, and RCs. Section 3 is dedicated to the description of headless RCs in Tigrinya. It first presents the morphosyntactic properties of LHRs, then turns to SFRs, and finally discusses free-choice RCs. Section 4 concludes.

2 Background on Tigrinya

2.1 Grammatical Properties

Tigrinya is an Ethio-Semitic language spoken in Eritrea and in the Tigray region in northern Ethiopia. It is closely related to languages such as Amharic and Tigré and more distantly related to Arabic and Hebrew.

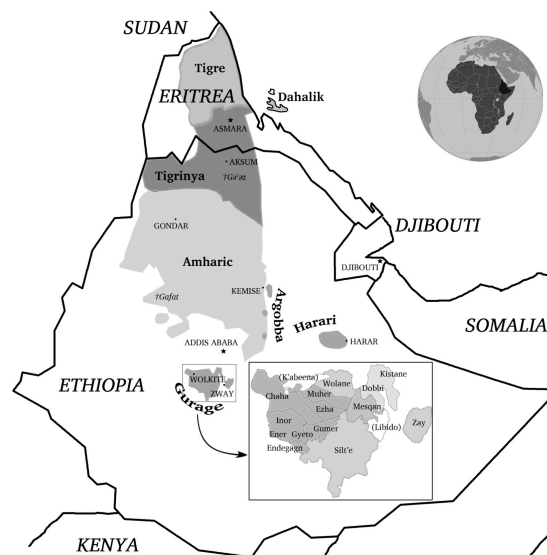


Figure 1: Distribution of Tigrinya speakers (adapted from The Semitic Languages ([Weninger \(2011\)](#)); credit: Ronny Meyer)

As with all the other Semitic languages, Tigrinya displays non-concatenative root morphology: words are formed from a combination of triconsonantal roots — that constitute the semantic core of the words — and vowel templates that are inserted into them. To form verbs, for example, vowel templates carry TAM (tense, aspect, modality) information. For instance, from the root /dqs/, which expresses the meaning of ‘sleep’, it is possible to obtain the three different simple aspects/tenses that are present in the language. The Imperfective is a prefixal conjugation — subject ϕ -features appear (mainly) as prefixes on the verb — and expresses habitual present tense, as in (4). The Perfective and the Gerundive are suffixal conjugations — subject ϕ -features appear as suffixes on the verb. The Gerundive, introduced through contact with Cushitic languages (Appleyard 2015), has replaced the Perfective in modern spoken language and expresses perfectivity (Bulakh 2019), as in (6). Despite the translation of (5) based on Nazareth 2011, the Perfective is only used in modern spoken Tigrinya as a suppletive form for the Gerundive in the presence of prefixes (cf. Bulakh 2019, Cacchioli 2023b, 2026).

- | | | |
|---|--|--|
| (4) ji-dɛqis
SM.3MS-sleep.IPFV
‘he sleeps’ | (5) dɛqɛs-ɛ
sleep.PFV-SM.3MS
‘he has slept’ | (6) dɛqis-u
sleep.GER-SM.3MS
‘he slept’ |
|---|--|--|

Tigrinya is a head-final language with SOV word order, as shown in (7). It is subject *pro*-drop. In complex verb forms that express progressive aspect the verb is accompanied by an auxiliary that follows it, as in (8). Note that both the verb and the auxiliaries carry subject marking morphology. Subordinate clauses precede matrix clauses, as in (9). The element that introduces complement clauses, *kɛmz(i)*-, has been labeled a conjunction (Leslau 1941, Mason 1996) or a complementizer (Tsfay 2016).²

- (7) ha-nti dimu riʔij-ɛ
 one.FS cat.FS see.GER-SM.1S
 ‘I saw a cat.’
- (8) ni-dɛrrif ʔall-ɛna
 SM.1P-sing.IPFV AUX.PRES-SM.1P
 ‘We are singing.’
- (9) [kullu gizie nab sinɛma kɛmz-i-xɛjjid] ti-fɛllit
 all time DIR cinema COMP-SM.3MS-go.IPFV SM.3FS-know.IPFV
 ‘She knows that he always goes to the cinema.’

Tigrinya aligns for Nominative-Accusative cases. Nominative case is unmarked with both transitive and unaccusative verbs, as shown in (10).

- (10) a. ʔit-i bazo tɛ-sɛbir-u
 DIST-MS vase.MS PASS-break.PFV-SM.3FS
 ‘The vase broke.’
- b. Tɛsfai n-it-i bazo sɛbir-u-wo
 Tsfay DOM-DIST-MS vase break.GER-SM.3MS-OM.3MS
 ‘Tsfay broke the vase.’

In contrast to Nominative case, Accusative case is differentially marked (Bossong 1991) with the prefix *n(i)*-, as in the demonstrative *ʔiti* ‘the’ that modifies *bazo* ‘vase’ in (10b).³ An object is marked with *n(i)*- only in a sentence in which that element is definite or specific (Weldeyesus

²*Kɛmz(i)*- has been claimed to be a bimorphemic element composed of the particle *kɛm* and the relative marker *z(i)*- (Cacchioli 2023a, 2026, van Urk 2024, 2025).

³The *n(i)*- prefix is also used to mark Dative case, encompassing “a wide range of adverbial meanings, in-

2004, Nazareth 2011, Overfelt 2022). When an object is marked by *n(i)-*, it is obligatorily cross-referenced by a ϕ -agreeing suffixal object marker on the verb (Kievit & Kievit 2009, Overfelt 2022). These markers always follow subject agreement if it surfaces, as shown in (11).

- (11) a. Təsfaj səbʔay qətil-u
Təsfaj man kill.GER-SM.3MS
'Təsfaj killed a man.'
- b. Təsfaj **n-it-i** səbʔay qətil-w-**o**
Təsfaj **DOM-DIST-MS** man kill.GER-SM.3MS-**OM.3MS**
'Təsfaj killed the man.'
- c. *Təsfaj səbʔay qətil-w-**o**
Təsfaj man kill.GER-SM.3MS-**OM.3MS**
'Təsfaj killed a man.'
- d. *Təsfaj **n-it-i** səbʔay qətil-u
Təsfaj **DOM-DIST-MS** man kill.GER-SM.3MS
'Təsfaj killed the man.'

In Tigrinya, nouns are inflected for number and gender. Singular nouns are unmarked, whereas plural nouns are marked either by a suffix attached to the noun — *-tat* (Mason 1996:10) — or by a noun-internal phonological change known in Semitic literature as *Broken Plural* (Palmer 1955). (12) shows a noun with a suffixal plural and a noun with broken plural.

- (12) a. dimu, **dimutat**
'cat, cats'
- b. komadara, **komidara**
'tomato, tomatoes'

There are two genders in Tigrinya: masculine and feminine. Animate nouns have inherent semantic gender, whereas inanimate objects do not have a stable grammatical gender (Mason 1996, Kogan 1997, Nazareth 2011).

Tigrinya has two sets of demonstrative determiners, always preceding their noun complement. Proximal demonstratives can only be used by the speaker to point at something close to them (Kogan 1997), as shown in (13).

- (13) **ʔiz-i** wədi, **ʔiz-a** g^wal, **ʔiz-om** ʔawədat, **ʔiz-en** ʔawəlɪd
PROX-MS boy, PROX-FS girl, PROX-MP boys, PROX-FP girls
'this boy, this girl, these boys, these girls'

Distal demonstratives, instead, exhibit a double behavior. Depending on the context of use, they can be interpreted as regular distal demonstratives pointing at something away from the speaker or in a way resembling the definite determiner in English — without any deictic component, but with an added uniqueness requirement. As a reminder, while *that boy* is preferable in a context with more than one boy that is salient, as long as the speaker is pointing at one of them, *the boy* can only be used in a context in which there is only one salient boy. The relevant Tigrinya examples are given in (14).

- (14) **ʔit-i** wədi, **ʔit-a** g^wal, **ʔit-om** ʔawədat, **ʔit-en** ʔawəlɪd
DIST-MS boy, DIST-FS girl, DIST-MP boys, DIST-FP girls
'the/that boy, the/that girl, the/those boys, the/those girls'

cluding direction, purpose, addressee" (Kievit & Kievit 2009:56). Henceforth, we gloss it as DOM, for Differential Object Marking, when marking accusative case; and as DAT when marking dative case.

Demonstratives of either kind can occur on their own without a noun complement and form DPs on their own, similarly to pronouns, as in (15) and (16).

- (15) ?iz-i/?it-i dɛlji-ɛ
 PROX-MS/DIST-MS want.GER-SM.1S
 ‘I want this/that.’

- (16) ?it-i!
 DIST-MS
 ‘That one!’

(from Nazareth 2011:29,(20c))

Tigrinya does not have a special marker to form a specific indefinite. A bare nominal can receive a definite or indefinite interpretation, according to the context, as in (18). Nevertheless, it is possible to use the numeral *hade* ‘one’ (*hanti* for feminine nouns) as a marker of specificity.

Adjectives are inflected according to number and gender of the noun they modify. In Tigrinya, they always occur between the demonstrative and the noun. In contrast, cardinal numerals in Tigrinya are not inflected for gender except for the first numeral ‘one’ just mentioned. When numerals occur in a phrase along with adjectives, they always precede the adjectives.

To sum up, demonstratives, numerals and adjectives in Tigrinya appear before the qualified noun in exactly this order. Examples (17) illustrates a sentence containing all these elements. Note that nouns modified by numerals and adjectives can also appear bare, namely without the demonstrative and receive an indefinite interpretation, as shown in (18).

- (17) ?it-om hammuftɛ ?abɛjti ?axlabat bunawi ?i-jjom
 DIST-MP five big dog.MP brown BE.PRES-SM.3MP
 ‘The five big dogs are brown.’

- (18) hammuftɛ ?abɛjti ?axlabat gɛza ?atij-om
 five big dog.MP house enter.GER-SM.3MP
 ‘Five big dogs entered the house.’

With regard to strong quantifiers such as *kull* ‘all’ and *mibizahiti?om* ‘most’, they agree in gender and number with the noun they modify, and their position has interpretive effects, as illustrated in (19) for *kull* ‘all’. If they are placed at phrase-initially, as in (19a), they scope over the whole phrase; if they appear after a determiner, as in (19b), they express a partitive reading (Nazareth 2011:33); if they occur at the end of the nominal phrase, as in (19d), they are emphasized and behave similarly to a floating quantifier. Strong quantifiers cannot appear after adjectives, as shown in (19c).

- (19) a. **kull-ɛn** (?it-ɛn) niwahi-ti dɛbdabɛ-tat
 all-FP DIST-FP long-F letter.F-P
 ‘all (the) long letters’
 b. ?it-ɛn **kull-ɛn** niwahi-ti dɛbdabɛ-tat
 DIST-FP all-FP long-F letter.F-P
 ‘all of the long letters’
 c. *(?it-ɛn) niwahi-ti **kull-ɛn** dɛbdabɛ-tat
 DIST-FP long-F all-FP letter.F-P
 d. (?it-ɛn) niwahi-ti dɛbdabɛ-tat **kull-ɛn**
 DIST-FP long-F letter.F-P all-FP
 ‘(the) long letters, all of them’

Weak quantifiers such as *bizuhat* ‘many’ and *wihudat* ‘few’ show adjectival inflection and their position also have interpretive effects. However, their distributional options seem to be more

restricted than strong quantifiers, as illustrated in (20). A fine-grained description and analysis of quantifiers in Tigrinya is outside the scope of this paper and is left to future research.

- (20) a. **bizuḥ-at** (*ʔit-ən) niwaḥ-ti dəbdabɛ-tat
many-FP DIST-FP long-F letter.F-P
‘(the) many long letters’
b. ʔit-ən **bizuḥ-at** niwaḥ-ti dəbdabɛ-tat
DIST-FP many-FP long-F letter.F-P
‘many of the long letters’
c. *(ʔit-ən) niwaḥ-ti **bizuḥ-at** dəbdabɛ-tat
DIST-FP long-F many-FP letter.F-P
d. ʔ(ʔit-ən) niwaḥ-ti dəbdabɛ-tat **bizuḥ-at**
DIST-FP long-F letter.F-P many-FP
‘(the) long letters, many of them’

Let us now move to interrogative clauses.

2.2 Interrogative Clauses

Tigrinya has several *wh*-elements, listed in Table 1.

<i>mən</i>	who
<i>ʔintaj</i>	what
<i>ʔajɛnaj</i>	which
<i>mɛʕas</i>	when
<i>ʔabɛj</i>	where
<i>kɛmɛj</i>	how
<i>nimintaj / silemintaj</i>	why

Table 1: *Wh*-elements in Tigrinya.

These elements are used to form interrogative *wh*-clauses, such as those in (21) and (22). Note that the *wh*-elements are in-situ.⁴

- (21) a. nisxa **ʔintay** bɛliʕi-ka
you.MS what eat.GER-SM.2MS
‘What did you eat?’
(22) a. nisxa **ni-mən** riʔi-ka-yo
you.MS DOM-who see.GER-SM.2MS-OM.3MS
‘Who did you see?’

Wh-elements are also used in embedded interrogative *wh*-clauses, such as those in (23) and (24), including complex embedded interrogative *wh*-clauses, as the one in (25). Note that embedded interrogative clauses are introduced by *kɛmz(i)*-. This aforementioned complementizer occurs in (23) because it is triggered by subordinate clauses introduced by different classes of verbs among which verbs of utterance such as ‘say’, ‘report’, ‘tell’ and ‘ask’.

⁴According to Gebregziabher & Duguine 2024, Tigrinya *wh*-elements may either remain in situ or move to the left periphery. This may reflect dialectal variation between our Eritrean consultants and the Ethiopian background of Gebregziabher, or alternatively involve syntactic mechanisms other than *wh*-movement (Cacchioli & Overfelt under review).

- (23) ʔaman [ʔintay kəmz-i-fəttu] hatit-u-ni
 Aman what COMP-SM.1S-like.IPFV ask.GER-SM.3MS-OM.1S
 ‘Aman asked me what I like.’
- (24) mɛʕas [(nissu) nab bet timihirti kəmz-i-xɛjjid] ninɛbs-ɛi
 when he to school COMP-SM.3MS-go.IPFV self-POSS.1S
 hatit-ɛ-ja
 ask.GER-SM.1S-OM.3FS
 ‘I wondered (lit. asked myself) when he goes to school.’
- (25) [ʔajən-ot mɛha z-ot nay ha ft-ɛj nab-t-i g^wajla
 which-M.P friend.P of sister-POSS.1S DIR-DIST-MS party
 kəmzi-mɛts-u] ʔaj-fɛlɛti-ku-n
 COMP-come.PFV-SM.3MP NEG-know.PFV-SM.1S-NEG
 ‘I didn’t know which friends of my sister came to the party.’

Next we turn to headed RCs.

2.3 Headed Relative Clauses

Headed RCs in Tigrinya are introduced by the relativizer *z(i)-*.⁵ Prenominal restrictive headed RCs occur between the demonstrative and the relativized noun, as illustrated below.

- (26) *Object Headed RC*
 [ʔit-a — lomi nigihō z-i-gɛzaʔi-ka-ja sofa]
 DIST-FS today morning REL-buy.PFV-SM.2MS-OM.3FS couch.FS
 ji-dɛli allɛ-xu
 SM.1S-want.IPFV BE.PRES-SM.1S
 ‘I want the couch that you bought this morning.’
- (27) *Subject Headed RC*
 [ʔit-i — məkajən z-i-ʃɛjɛt sɛb] haw-ɛj
 DIST-MS car.P REL-SM.3MS-sell.IPFV man brother-POSS.3MS
 ʔi-jju
 be.PRES-SM.3MS
 ‘The man who sells cars is my brother.’

None of the *wh*-elements occurring in interrogative clauses can ever occur in headed RCs, as shown in (28) for the *wh*-element *ʔintaj* ‘what’. The complementizer *kəmz(i)-* is also ungrammatical in headed RCs, as shown in (29).

⁵The vowel [i] is epenthetic; the morpheme itself is *z*. Tigrinya grammars report that *z(i)-* surfaces as *ʔ(i)-* when prefixed to a [t] phoneme of the Imperfective second person and third feminine singular person, and the [n] phoneme of first person plural (Leslau 1941, Mason 1996, Nazareth 2011, Bulakh 2019). However, our consultants do not produce *ʔ(i)-* (either when speaking or when writing) and instead leave the verb bare. We chose examples in which *z(i)-* is clearly recognizable.

(28) *Object Headed RC with a wh-element*

*[?it-a **?intaj** lomi nigihō **zi-geza?i-ka-ja** **sofa**]
DIST-FS **what** today morning REL-buy.PFV-SM.2MS-OM.3FS couch.FS
ji-deli alle-xu
SM.1S-want.IPFV BE.PRES-SM.1S

Intended: ‘I want the couch that you bought this morning.’

(29) *Object Headed RC with kemz(i)-*

*[?it-a ____ lomi nigihō **kemzi-geza?i-ka-ja** **sofa**]
DIST-FS today morning COMP-buy.PFV-SM.2MS-OM.3FS couch.FS
ji-deli alle-xu
SM.1S-want.IPFV BE.PRES-SM.1S

Intended: ‘I want the couch that you bought this morning.’

Three further observations will turn out to be useful for the discussion of headless RCs later. First, postnominal headed RCs are attested in Tigrinya, as shown in (30). However, such headed RCs are only marginally grammatical in this position (Palmer 1962), and the possible semantic differences between postnominal and prenominal headed RCs remain to be investigated.

- (30) a. [?it-a ____ zi-re?a-ku-ja **ʁuf**] t’efi?-a
DIST-FS REL-see.PFV-SM.1S-OM.3FS bird.FS disappear.GER-SM.3FS
b. [?it-a **ʁuf** zi-re?a-ku-ja] t’efi?-a
DIST-FS bird.FS REL-see.PFV-SM.1S-OM.3FS disappear.GER-SM.3FS
‘The bird that I saw has disappeared.’

Second, unlike subject headed RCs, such as the one in (27), object headed RCs, such as the one in (26), require the relativized verb to carry an object marker that agrees in number, person, and gender with the nominal head. This explains the ungrammaticality of (31), in which the masculine object marker *-(j)o* fails to agree with the feminine nominal head *sofa* ‘couch’.⁶

- (31) [*?it-a ____ lomi nigihō zi-geza?i-ka-**jo** sofa]
DIST-FS today morning REL-buy.PFV-SM.2MS-OM.3MS couch.FS
ji-deli alle-xu
SM.1S-want.IPFV BE.PRES-SM.1S
‘I want the couch that you bought this morning.’ (Lit.: ‘I am wanting...’)

Third, as in other Semitic languages such as Standard Arabic and Hebrew, relativization of a prepositional complement is realized through the appearance of a resumptive pronoun attached to the preposition, as illustrated in (32).

- (32) [?it-a ____ timali mifet **məs-*(?a)** zi-saʕisaʕi-ku-wa g^wal]
DIST-FS yesterday night with-OM.3FS REL-dance.PFV-SM.1S-OM.3FS girl
ji-fitti-wa
SM.1S-like.IPFV
‘I like the girl with whom I danced yesterday night.’

We assumed *z(i)-* to be a relativizer, following various grammars (a.o. Leslau 1941, Mason 1996, Kogan 1997). Recent work focusing specifically on *z(i)-* (Cacchioli 2023b, 2026) has argued that this element should be better analyzed as a reflex of successive-cyclic $\bar{A}$ -movement, since it can also occur in constructions other than headed RCs that, like headed RCs, involve

⁶The occurrence of the glide [j] is due to phonological processes that are beyond the scope of this paper.

this type of movement, as shown in (33) with comparative clauses and in (34) with noun clausal complements.

(33) *Comparative Clauses*

Selam [kab-t-i **zi**-hasəb-ki-wo] nilaŋili məšhar
 Selam.FS from-DIST-MS **REL**-think.PFV-SM.1S-OM.3MS more funny
 ʔi-jja
 BE.PRES-SM.3FS
 ‘Selam is funnier than I thought.’

(34) *Noun Clausal Complements*

ʔit-i [Solomon feres geziʔ-u **z-i**-bił] zərəba hasot
 DIST-MS Solomon horse buy.GER-SM.3MS **REL**-SM.3MS-say.IPFV story.MS false
 ʔi-jju
 COP.PRES-SM.3MS
 ‘The story that Solomon bought a horse is false.’

We refer the reader to [Overfelt 2009](#), [Cacchioli 2023b](#), [2026](#) for further details and morphosyntactic analyses of headed RCs in Tigrinya.

Next we turn to headless RCs, the family of constructions at the center of our investigation.

3 Headless Relative Clauses

In this section, we present headless RCs in Tigrinya in detail, starting with LHRs (Section 3.1), continuing with SFRs (Section 3.2), and concluding with a variety of headless RCs that we have not introduced yet: *free choice headless relative clauses* (Section 3.3). The emerging picture is that all headless RCs in Tigrinya rely on the relativization strategy using the same relative marker *z(i)-* as the one in headed RCs, while none of them makes use of *wh*-elements nor the complementizer *kəmz(i)-* that occur in interrogative clauses.

3.1 Light-Headed Relative Clauses

LHRs lack a nominal head, like all headless RCs, and are characterized by a “lighter” D head that combines with them ([Citko 2004](#), [Caponigro 2021](#), [2023](#)). In Tigrinya, the D heads that can introduce LHRs are demonstratives (Section 3.1.1), numerals (Section 3.1.2), and quantifiers (Section 3.1.3), while pronouns or *wh*-elements can never introduce LHRs (Section 3.1.4).

3.1.1 LHRs with Demonstratives

As mentioned in Section 2.1, distal demonstratives also function as definite determiners in the appropriate context, when combined with a noun. When combined with an RC to form an LHR, a distal demonstrative always loses its deictic features and only behaves like a definite determiner. It has been observed before that demonstratives introducing headed or headless RCs often behave like definite determiners, rather than deictic demonstratives ([King 2001](#), [Citko 2004](#), [Simonenko 2014](#), [Mantenuto & Caponigro 2021](#), [Ahn 2022](#)). Examples of LHRs introduced by distal demonstratives are given in (35) and (36).

- (35) [**?it-i** — **z-i-ʃet'-o**] k-i-gɛzi?
 DIST-MS REL-SM.3MS-sell.IPFV-OM.3MS SUBJ-SM.1S-buy.IPFV
 ?i-jje
 COP.PRES-SM.3MS
 'I will buy the stuff he sells.'
- (36) [**?it-i** — **?ab-z-i** bet mɛgbi **z-i-bɛlʃ-o**] t'iʃum
 DIST-MS LOC-PROX-MS restaurant REL-SM.3MS-eat.IPFV-OM.3MS delicious
 ?i-jju
 COP.PRES-SM.3MS
 'The stuff I eat in this restaurant is delicious.'

As our translations into English make it clear, the bracketed LHRs in (35) and (36) are interpreted as definites. For instance, the LHR in (35) can only mean 'the stuff/things he sells'. It cannot be interpreted as 'that stuff he sells that is close to the speaker and the speaker is pointing at' (in contrast with some other stuff that he sells that is not close to the speaker or the speaker is not pointing at). Notice that the loss of deictic features with the distal demonstrative when introducing an LHR is observed in English as well, as shown in (37).

- (37) I interacted with [**those** who were sitting near me at dinner].

The bracketed LHR is introduced by the distal demonstrative *those* but refers to all the people who were sitting near the speaker rather than a subgroup of people that are far away from the speaker and the speaker is pointing at. A similar pattern is attested in LHRs in Teramano (Mantenuto & Caponigro 2021:sec. 3.2).

In (35) and (36), the distal demonstrative occurs in its masculine singular form. This is the only possible form when the LHR refers to unspecified, general things (e.g., the things somebody sells or the things the speaker eats). The use of the feminine singular form or the plural form is unacceptable, as shown in (38) and (39), respectively.

- (38) [***?it-a** — **z-i-ʃet'-o**] k-i-gɛzi?
 DIST-FS REL-SM.3MS-sell.IPFV-OM.3MS SUBJ-SM.1S-buy.IPFV
 ?i-jje
 COP.PRES-SM.3MS
 'I will buy what he sells.'
- (39) [***?it-om** — **z-i-ʃet'-o**] k-i-gɛzi?
 DIST-MP REL-SM.3MS-sell.IPFV-OM.3MS SUBJ-SM.1S-buy.IPFV
 ?i-jje
 COP.PRES-SM.3MS
 'I will buy what he sells.'

On the other hand, if the context makes it clear that we are talking about things that are labeled with a feminine noun and the LHR refers to one of them, then the singular feminine demonstrative has to be used to introduced the LHR, as shown in (40).

- (40) [**?it-a** — **zi-dɛlɛ-ka-ja**] ki-ti-bɛlʃ
 DIST-FS REL-want.PFV-SM.2MS-OM.3FS SUBJ-SM.2MS-eat.IPFV
 ti-xiʔil
 SM.2MS-can.IPFV
 'You can eat the one that you want.'

Prepositions cliticize on demonstratives, but always on the left, as shown in (41) and (42).

- (41) **?ab-** [**t-i** qolʃu — z-i-ts'awet-u-lu] ʃigara
 LOC- -DIST-MS children REL-SM.3PM-play.IPFV-SM.3MP-OBL-3MP cigarette
 ki-ti-tikik ʔaj-ti-xiʔili-n
 SUBJ-SM.2MS-smoke.IPFV NEG-SM.2MS-can.IPFV-NEG
 'You can't smoke a cigarette where the children play.'
- (42) **nab-** [**t-i** — z-i-xejjid-o] ji-xejjid
 DIR- DIST-MS REL-SM.3MS-go.IPFV SM.1S-go.IPFV
 'I go where he goes.'

A preposition cannot cliticize to the right, intervening between the light head and the RC, as shown by the contrast between (43a) and (43b) .

- (43) a. **nab-** [**t-i** — zi-teweldi-ku-lu] keid-ε
 DIR DIST-MS REL-be_born.PFV-SM.1S-OBL.3MS go.GER-SM.1S
 bet timihirit-ay riʔij-Ø-a
 school-POSS.1S see.GER-SM.1S-OM.3FS
 b. [***ʔit-i nab** — zi-teweldi-ku-lu] keid-ε
 DIST-MS DIR REL-be_born.PFV-SM.1S-OBL.3MS go.GER-SM.1S
 bet timihirit-ay riʔij-Ø-a
 school-POSS.1S see.GER-SM.1S-OM.3FS
 'I went where I was born and saw my school.'

The contrast in (43) can be taken as evidence that the preposition is external to the LHR and takes the LHR as its complement, rather than being inside the LHR — a conclusion we highlight in (43a) with the position of the open square bracket. Further evidence in favor of this conclusion comes from selection. The directional preposition in (43a) is selected by the matrix predicate ('go'), while the LHR predicate ('be born') license a locative PP adjunct. The variant with a locative preposition instead of directional preposition is unacceptable, as shown in (44).

- (44) [***ʔab-** t-i — zi-teweldi-ku-lu] keid-ε
 LOC- DIST-MS REL-be_born.PFV-SM.1S-OBL.3MS go.GER-SM.1S
 bet timihirit-ay riʔij-Ø-a
 school-POSS.1S see.GER-SM.1S-OM.3FS
 'I went where I was born and saw my school.'

Like definite nominals, LHRs with a demonstrative D head are interpreted as referring to a specific thing or to something generic depending on the interpretation of the whole sentence. For instance, (45) is interpreted as a specific statement about the speaker eating at the time of utterance the specific food they prepared one hour earlier because of the present tense marking in the matrix clause and temporal adverb 'one hour ago' in the LHR. Therefore, the LHR in (45) is interpreted as referring to some specific food. By contrast, (46) is interpreted as a generic statement because of the imperfective marking on the matrix predicate and the temporal adverbial quantifier 'everyday'. Therefore, the LHR in (46) does not refer to some specific food but its reference varies day by day.

- (45) [ʔit-i — **qidmi ha-de saʃat** zi-dali-na-jo]
 DIST-MS **ago one.MS hour** REL-prepare.PFV-SM.1P-OM.3MS
 ni-belʃ-jo ʔallε-na
 SM.1P-eat.IPFV-OM.3MS AUX.PRES-SM.1P
 'We are eating what we prepared **one hour ago**.'

- (46) [?it-i — **meʕaltawi** zi-dali-na-jo]
 DIST-MS **every_day** REL-prepare.PFV-SM.1P-OM.3MS
 ni-belʕ-jə
 SM.1P-eat.IPFV-OM.3MS
 ‘Everyday we eat what we prepared.’

Note also that LHRs are grammatical in the presence of an adjective that follows the light head, as illustrated in (47).

- (47) [?it-i **kibur** — ?ab-z-i bet migbi z-i-belʕ-o]
 DIST-MS **expensive** LOC-PROX-MS restaurant REL-SM.1S-eat.IPFV-OM.3MS
 tʕiʕum ?i-jju
 delicious COP.PRES-SM.3MS
 ‘The expensive (food) I eat in this restaurant is delicious.’

Proximal demonstratives too can introduce LHRs in Tigrinya. Unlike distal demonstratives, though, proximal demonstratives do not lose their deictic features. Therefore, the whole LHR inherits those. For instance, (48a) is a familiar LHR introduced by a distal demonstrative and just refers to all the things the speakers prepared, regardless whether they are close or far from the speakers. By contrast, the LHR in (48b) is introduced by a proximal demonstrative and refers to the things that the speakers prepared and are closed to them, excluding things that they prepared and are far of them. Similarly, the LHR in (49a) is introduced by a distal demonstrative and refers to all the stuff that somebody contextually salient makes. By contrast (49b) contains an LHR introduced by a proximal demonstrative that only refers to the stuff that the contextually salient individual makes and the speaker can see or is at least in close proximity to.

- (48) a. [?it-i — zi-dali-na-jo] ni-belʕ-jə
 DIST-MS REL-prepare.PFV-SM.1P-OM.3MS SM.1P-eat.IPFV-OM.3MS
 ‘We eat the stuff we prepared.’
 b. [?iz-i — zi-dali-na-jo] ni-belʕ-jə
 PROX-MS REL-prepare.PFV-SM.1P-OM.3MS SM.1P-eat.IPFV-OM.3MS
 ‘We eat the stuff over here we prepared.’
- (49) a. [?it-i — z-i-gebbir-o] tʕeqami ?i-jju
 DIST-MS REL-SM.3MS-do.IPFV-OM.3MS useful COP.PRES-SM.3MS
 ‘The stuff he makes is useful.’
 b. [?iz-i — z-i-gebbir-o] tʕeqami ?i-jju
 PROX-MS REL-SM.3MS-do.IPFV-OM.3MS useful COP.PRES-SM.3MS
 ‘This stuff over here he makes is useful.’

This contrast between the behavior of distal vs. proximal demonstratives in introducing LHRs is attested in Teramano (Italo-Romance) as well, with a third deictic dimension due to the presence of medial demonstratives (Mantenuto & Caponigro 2021:sec. 3.2).

All the examples of LHRs we have discussed have a corresponding grammatical version without the demonstrative light head. These would be all examples of SFRs, which we present and discuss later, in Section 3.2.

3.1.2 LHRs with Numerals

LHRs in Tigrinya can also be introduced by cardinal numerals (henceforth, simply “numerals”). Examples are given in (50) and (51). Contexts help identify the reference of LHRs introduced by numerals, but they are not always necessary.

- (50) *Context: A police officer tells a passer-by that he is looking for some men who started pushing each other in a bar and then ran out, escalating the situation. The passer-by replies:*

[**kiltə** ____ ʔab godəna zi-qaləs-u] riʔij-ε
two LOC streets RELfight.PFV-SM.3MP see.GER-SM.1S

‘I saw two fighting in the streets!’⁷

- (51) *Context: Two children are talking. The first exclaims that she loves books; the second replies:*

ʔanə wun! [**hanti** ____ ʔazijə z-i-fetu-wa] Harry Potter
 me too **one.F** really REL-SM.1S-like.IPFV-OM.3FS Harry Potter
 ʔi-jja
 COP.PRES-SM.3FS

‘Me too! One that I really like is *Harry Potter*.’

Numerals can co-occur with demonstratives when introducing LHRs. In this case, the demonstrative always occurs first, followed by the quantifier or the RC, as shown in (52) and (53) which are counterparts of (50) and (51). By adding a demonstrative, the whole DP turns into a definite, as highlighted by the English translation. For example, (51) states that there exists at least one book that the speaker really likes and *Harry Potter* is one such book; however, (53) expresses that there is exactly one relevant book that the speaker really likes, and it is *Harry Potter*.

- (52) a. [**ʔit-om kiltə** ____ ʔab godəna zi-qaləs-u] riʔij-ε
 DEM-MP **two** LOC streets RELfight.PFV-SM.3MP see.GER-SM.1S
 b. * [**ʔit-om** ____ ʔab godəna zi-qaləs-u **kiltə**] riʔij-ε
 DEM-MP LOC streets RELfight.PFV-SM.3MP **two** see.GER-SM.1S
 ‘I saw the two fighting in the streets!’

- (53) ʔanə wun! [**ʔit-a hanti** ____ ʔazijə z-i-fetu-wa] Harry Potter
 me too DEM-FS **one.F** really REL-SM.1S-like.IPFV-OM.3FS Harry Potter
 ʔi-jja
 COP.PRES-SM.3FS
 ‘Me too! The one that I really like is *Harry Potter*.’

3.1.3 LHRs with Quantifiers

LHRs can also have quantifiers other than numerals act as their D heads in Tigrinya. A quantifier light head can occur at the very beginning or at the very end of its LHR, as shown in (54) and (55). This distributional behavior resembles the one that the same quantifiers exhibit when they combine with a simple nominal, as detailed in Section 2.1 and mentioned again in Section 3.1.2.

⁷Note that the light head *kiltə* ‘two’ can also be placed after the *z(i)*-marked verb. In this case, it seems that the light head is emphasized and behaves like a floating quantifier, as explained for *kull* ‘all’ in example (19).

- (54) a. [**bizuḥ** ____ zi-beli-ku-ka] ts'ihif-ε-jo
a_lot REL-say.PFV-SM.1S-OM.2MS write.GER-SM.1S-OM.3MS
b. [____ zi-beli-ku-ka **bizuḥ**] ts'ihif-ε-(*jo)
REL-say.PFV-SM.1S-OM.2MS a_lot write.GER-SM.1S-OM.3MS
'I wrote down a lot of what I told you.'⁸

(55) Context: *I entered a dark restaurant and...*

- a. [**bizuḥat** ḥadεgepa-tat ____ z-i-mesl-u] reʔij-ε
many dangerous-P REL-SM.3MP-seem.IPFV-SM.3MP see.GER-SM.1S
b. [ḥadεgepa-tat ____ z-i-mesl-u **bizuḥat**] reʔij-ε
dangerous-P REL-SM.3MP-seem.IPFV-SM.3MP many see.GER-SM.1S
'I saw many who seemed dangerous.'

Quantifiers such as *bizuḥ* 'a lot' can also co-occur with demonstratives when introducing LHRs, with no apparent change in meaning, as shown in (56). For one of our consultants, the presence of the demonstrative is, in fact, preferred.

- (56) a. [**ʔit-i** **bizuḥ** ____ zi-beli-ku-ka] ts'ihif-ε-jo
DIST-MS a_lot REL-say.PFV-SM.1S-OM.2MS write.GER-SM.1S-OM.3MS
b. [**ʔit-i** ____ zi-beli-ku-ka **bizuḥ**] ts'ihif-ε-jo
DIST-MS REL-say.PFV-SM.1S-OM.2MS a_lot write.GER-SM.1S-OM.3MS
'I wrote down a lot of what I told you.'

With regard to the quantifier *kull* 'all', the demonstrative is obligatory.⁹

- (57) a. [***(ʔit-om)** **kull-om** ____ zi-goje-j-u] qelt'if-om mets'iʔ-om
DIST-MP all-MP REL-run.PFV-SM.3MP quick-MP come.GER-SM.3MP
b. [***(ʔit-om)** ____ zi-goje-j-u **kull-om**] qelt'if-om mets'iʔ-om
DIST-MP REL-run.PFV-SM.3MP all-MP quick-MP come.GER-SM.3MP
'All those who ran came quickly.'

We have seen that LHRs can be introduced by determiners, demonstratives, numerals, and quantifiers. In the next section, we focus on D elements that cannot introduce LHRs in Tigrinya, although they can in other languages.

3.1.4 No LHRs with pronouns or *wh*-elements as light heads

Personal pronouns can behave like "light" D heads and introduce LHRs in English (although the whole LHR may sound archaic or idiom-like), as shown in (58).

- (58) [**He** who laughs last], laughs best.

In Tigrinya, instead, personal pronouns can never introduce LHRs. The LHR in (59) is introduced by a distal demonstrative, an option that is fully productive (see Section 3.1.1). If the demonstrative is replaced with a personal pronoun, the result is unacceptable, regardless of whether the personal pronoun precedes or follows the RC, as shown in (60).

⁸Why the object marker *-(j)o* is ungrammatical when the quantifier *bizuḥ* 'a lot' appears after the relativized verb is left to future research.

⁹In Section 2.1, we showed that *kull* 'all' can occur without a demonstrative when it combines with a simple noun. The nature of this contrast deserves further investigation.

- (59) [**?it-i** — ?ab mɛndɛq zi-ts' ɛhaf-ɛ] hadim-u
 DIST-MS LOC wall REL-write.PFV-SM.3MS flee.GER-SM.3MS
 ‘The one who wrote on the wall ran away.’
- (60) [***nissu** — ?ab mɛndɛq zi-ts' ɛhaf-ɛ] hadim-u
he LOC wall REL-write.PFV-SM.3MS flee.GER-SM.3MS
 (‘The one who wrote on the wall ran away.’)

The acceptable example in (61) shows a personal pronoun followed by an LHR introduced by a demonstrative. Crucially, a pause is required between the pronoun and the LHR, signaling that the LHR behaves like a parenthetical and that it does not form a construction with the pronoun.

- (61) **nissu**, [?it-i — ?ab mɛndɛq zi-ts' ɛhaf-ɛ,] hadim-u
he DIST-MS LOC wall REL-write.PFV-SM.3MS flee.GER-SM.3MS
 ‘**He**, the one who wrote on the wall, ran away.’

Before moving to headless RCs that lack even a light head, let us consider *wh*-elements again. In languages like English or Polish, *wh*-elements can introduce headless RCs on their own, resulting in a free relative clause, like the one in (62):

- (62) [**What** is on the table] is mine.

Or they can co-occur with light heads like pronouns (see ex. (58)) or demonstratives, as in (63):

- (63) [Those **who** work carefully and deeply] publish less.

While in English some *wh*-elements can occur in headed RCs, in Polish no *wh*-elements can occur in headed RCs (Citko 2004). In Tigrinya, instead, no *wh*-element can ever introduce any kind of RCs, regardless of whether they are headed (as already shown in (28)) or headless. In headless RCs, the *wh*-element cannot occur on its own, as shown in (64), or with a demonstrative, as shown in (65), nor with a pronoun acting as a light head, as shown in (66).

- (64) [*?abzi **mɛn** zi-nɛbɛr-ɛ] n-ɛt-a mɛts' haf wɛsid-ɛ
 here who REL-COP.PFV-SM.3MS DOM-DIST-FS book.FS take.GER-SM.3MS
 (‘The one who was here took the book.’)
- (65) [*?it-i **?intaj** zi-gebir-o] gega nɛjr-u
 DIST-MS what REL-do.PFV-SM.3MS wrong COP.GER-SM.3MS
 (‘What he did was wrong.’)
- (66) [*nissu **mɛn** ?ab mɛndɛq zi-ts' ɛhaf-ɛ] hadim-u
 he who LOC wall REL-write.PFV-SM.3MS flee.GER-SM.3MS
 (‘The one who wrote on the wall, he ran away.’)

We now turn to headless RCs that lack even a light head.

3.2 Super-Free Relative Clauses

In this section, we show that Tigrinya makes productive use of a kind of headless RC that has only started being investigated: *Super-Free Relative Clauses* (SFRs).¹⁰ SFRs are characterized crosslinguistically by the lack of a nominal head (like all headless RCs), the lack of a D element acting as a light head, and the lack of a *wh*-element. They can feature a complementizer, a

¹⁰The label “super-free relative clauses” is due to Caponigro et al. 2021, which also presents the first crosslinguistic description of SFRs, focusing on Mesoamerican languages. Sadler & Camilleri 2018 present SFRs in Maltese. Caponigro 2023 proposes a general syntactic and semantic analysis of SFRs across languages.

non-*wh* relativizer, or no marker at all (see Caponigro et al. 2021 for examples of each kind). Tigrinya makes use of the relativizer strategy: all its SFRs are characterized by the relativizer *z(i)-*, besides the usual missing constituent — a feature shared by all RCs in Tigrinya. Examples are given in (67) and (68).

Example (67) shows the relativization of the subject within the SFR, with the SFR itself being the subject of the matrix clause. (68) shows the relativization of the direct object of the SFR, with the SFR itself being the direct object of the matrix clause. In both SFRs, only the relative marker verbal prefix *z(i)-* characterizes them, together with the missing argument.

- (67) [____ *n-iz-i* *sirah zɪ-sɛɾɛh-Ø-a*] *ħariβu*
 DOM-PROX-MS job REL-work.PFV-SM.3MS-OM.3FS angry
 nɛir-u
 COP.GER-SM.3MS
 ‘The one who did this job was angry.’
- (68) [____ *z-i-ʃɛt'-o*] *k-i-gɛziʔ* *ʔi-jjɛ*
 REL-SM.3MS-sell.IPFV-OM.3MS SUBJ-SM.1S-buy.IPFV COP.PRES-SM.3MS
 ‘I will buy what he sells.’

Constituents other than subjects and direct objects are introduced by specialized markers. In particular, a directional preposition taking an NP complement is used for directional arguments, as shown in (69). When an SFR behaves like a directional argument of the matrix clause, it is introduced by the same directional preposition as in (69), in the same position as the NP complement, as shown in (70).

- (69) **nab** *gɛzi-ʔa* *ti-xɛjjid*
 DIR home-POSS.3FS SM.3FS-go.IPFV
 ‘She goes (to her) home.’
- (70) **nab** [____ *z-i-xɛjjid-o*] *ji-xɛjjid*
 DIR REL-SM.3MS-go.IPFV SM.1S-go.IPFV
 ‘I go where he goes.’

Note that the directional preposition immediately preceding the SFR has to satisfy the requirements of the matrix, rather than of the SFR itself. (71) shows an example of a headed RC in which the matrix predicate ‘go’ selects for a directional complement, while the SFR predicate ‘to be born’ requires a locative adjunct. The SFR has to be preceded by the directional preposition, as shown in (72a); if preceded by a locative preposition, the result is judged unacceptable, as shown in (72b).

- (71) Rut **nab-** [*t-i* *ʔaman zi-tiwɛlid-ɛ-lu* *bota*] *kɛjjid-a*
 Ruth DIR- DIST-MS Aman REL-be_born-SM.3MS-OBL.3MS place go.GER-SM.3FS
 ‘Ruth went to the place where Aman was born.’
- (72) a. Rut **nab** [*ʔaman zi-tiwɛlid-ɛ-lu*] *kɛjjid-a*
 Ruth DIR- Aman REL-be_born-SM.3MS-OBL.3MS go.GER-SM.3FS
 ‘Ruth went where Aman was born.’
- b. *Rut **ʔab** [*ʔaman zi-tiwɛlid-ɛ-lu*] *kɛjjid-a*
 Ruth LOC- Aman REL-be_born-SM.3MS-OBL.3MS go.GER-SM.3FS
 (‘Ruth went where Aman was born.’)

This contrast may be taken as showing that the preposition is external to the SFR and that it is taking the whole SFR as its complement. This conclusion matches the conclusion we reached for LHRs occurring with a preposition (Section 3.1.1).

Constituents like recipients, beneficiaries, and goals are marked by the dative marker *n(i)-*, which prefixes to the DP, as shown in (73). The dative marker is compatible with both definite and indefinite nominals. When an SFR occurs as a recipient, or a beneficiary, or a goal, then the dative marker is prefixed on the SFR predicate, as shown in (74). By contrast, when an LHR occurs as a recipient, or a beneficiary, or a goal, then the dative marker is prefixed on the light head, as shown in (75).

- (73) a. *n-it-i* *habereta* **ni-***Təsfay* *hab-o*
 DOM-DIST-MS information **DAT-***Təsfay* give.IMP-OM.3MS
 ‘Give the information to Təsfay!’
 b. *n-it-i* *habereta* **ni-hadε** *temiharaj* *hab-o*
 DOM-DIST-MS information **DAT-one.M** student give.IMP-OM.3MS
 ‘Give the information to a student!’
- (74) *Təsfaj n-it-i* *habereta* **ni-** [*zi-hatot-o*]
 Təsfaj DOM-DIST-MS information **DAT-** REL-ask.PFV-SM.3MS-OM.3MS
 hib-w-o
 give.GER-SM.3MS-OM.3MS
 ‘Təsfaj gave the information to whoever asked him.’¹¹
- (75) *Təsfaj *(n)-it-om* [*?aman ginzeβ* (***n)-zi-hab-Ø-om**] *ginzeβ*
 Təsfaj DAT-DIST.MP *Aman* money DAT-REL-give.PFV-SM.3MS-OM.3MP money
 hib-w-om
 give.GER-SM.3MS
 ‘Təsfaj gave money to those to whom Aman gave money.’

Since this prefix is most commonly found on nouns in Tigrinya, this may suggest that RCs in Tigrinya are nominal constructions — a possibility that has not been considered in the existing literature. This issue is left for future research.

3.2.1 Object markers

Like headed RCs and LHRs, SFRs exhibit object marking (as the right-most verbal suffix) when their direct object is relativized. The SFR in (68) above showed the familiar default object marking — third person masculine singular. If the context allows, third person feminine singular object marking can be used as well, as shown in (76).

¹¹The open square bracket (marking the beginning of the RC) is placed between *n(i)-* and the *z(i)-*marked verb. This is because the dative is selected by the matrix verb /hb/ ‘give’ and, therefore, should be part of the matrix clause, rather than the RC.

(76) *Context: A mother takes her child to a park with many small animals, such as dogs, cats, and sheep, all of which are labeled with feminine nouns in Tigrinya. She tells her child:*

- a. [____ zi-dɛɛ-ka-**jo**] ki-ti-tɛnikif
REL-want.PFV-SM.2MS-OM.3MS SUBJ-SM.2MS-touch.IPFV
ti-xiʔil
SM.2MS-can.IPFV
‘You can touch what(ever) you want.’
- b. [____ zi-dɛɛ-ka-**ja**] ki-ti-tɛnikif
REL-want.PFV-SM.2MS-OM.3FS SUBJ-SM.2MS-touch.IPFV
ti-xiʔil
SM.2MS-can.IPFV
‘You can touch whichever (*female* dogs, cats, and sheep around us) you want.’
- c. [*____ zi-dɛɛ-ka-**jom/jin**] ki-ti-tɛnikif
REL-want.PFV-SM.2MS-OM.3MP/3FP SUBJ-SM.2MS-touch.IPFV
ti-xiʔil
SM.2MS-can.IPFV
- d. [*____ zi-dɛɛ-ka-**Ø**] ki-ti-tɛnikif ti-xiʔil
REL-want.PFV-SM.2MS-Ø SUBJ-SM.2MS-touch.IPFV SM.2MS-can.IPFV

The SFR in (76a) with 3MS object marking is interpreted as referring to all the things in the context (animals, trees, flowers, fences, not just those that are animate and female). The SFR in (76b), instead, carries 2FS object marking and is interpreted as referring to the contextually salient female animals (i.e., dogs, cats, and sheep). The same SFR with plural object marking in (76c) is judged unacceptable, regardless of whether masculine or feminine. Finally, the SFR without any object marking in (76d) is judged unacceptable as well.

The only way to have a plural object marker is by adding a demonstrative, thereby turning the SFR into an LHR, as shown in (77).

- (77) [n-**it-in** ____ zi-dɛɛ-ka-**jɪn**] ki-ti-bɛɪɪ
DOM-DIST-FP REL-want.PFV-SM.2MS-OM.3FP SUBJ-SM.2MS-eat.IPFV
ti-xiʔil
SM.2MS-can.IPFV
‘You can eat those that you want.’

We now turn to existential SFRs.

3.2.2 Existential SFRs

The interpretation of headed RCs and LHRs crucially depends on their D head. If it is a demonstrative, then they will be interpreted as referential; if it is a quantifier or a numeral, then they will be interpreted as quantificational. SFRs, instead, are always interpreted as referential. For instance, the bracketed string in (78a) is an SFR referring to the things that the addressee cooked, which is exactly the same interpretation assigned to the bracketed string in (78b), which is an LHR introduced by a distal determiner.

- (78) a. [____ zi-bɛsɛɪ-ka-jo] tʰiʔim-ɛ-jo gin
REL-cook.PFV-SM.2MS-OM.3MS taste.GER-SM.1S-OM.3MS but
ʔaj-fiɛ-ku-wo-n
NEG-like.PFV-SM.1S-OM.3MS-NEG
‘I tasted what you cooked but I did not like it.’

- b. [**ʔit-i** — zi-bəsəl-ka-jo] t'ifim-ε-jo gin
 DIST-MS REL-cook.PFV-SM.2MS-OM.3MS taste.GER-SM.1S-OM.3MS but
 ʔaj-fitε-ku-wo-n
 NEG-like.PFV-SM.1S-OM.3MS-NEG
 'I tasted what you cooked but I did not like it.'

The only exception to SFRs being interpreted as referential is when SFRs occur as complements of a restricted class of existential or intensional predicates. In this case, they are interpreted as indefinites. Examples are provided in (79)–(81).

- (79) [— kullu gizie 'ʔiwε' z-i-bil-u] **ʔall-εwu**
 all time yes REL-SM.3MP-say.IPFV-SM.3MP COP.PRES-SM.3MP
 'There are people that always say 'yes'.'
 (80) [— ʔab-z-i fabrika k-i-sərrih z-i-dəli]
 LOC-PROX-MS factory SUBJ-SM.3MS-work.IPFV REL-SM.3MS-want.IPFV
j-ell-e-n
 NEG-COP.PRES-SM.3MS-NEG
 'There is nobody who wants to work in this factory.' (Lit.: 'There aren't people who want to work in this factory')
 (81) [— hadəga ʔintε-gatim-u zi-dəwəl-ε-lu]
 danger if-happen.PFV-SM.3MS REL-call.PFV-SM.3MS-OBL.3MS
ʔaj-neber-ε-n
 NEG-COP.PFV-SM.3MS-NEG
 'There was nobody to call in case of danger.' (Lit.: 'There weren't people to call in case of danger.')

The SFRs in (79)–(81) occur as the complement of existential copulas, roughly the equivalent of *there + be* in English, and are interpreted as indefinites, as shown by the use of bare plurals in the English translations as the heads of the headed RCs that are used to render Tigrinya SFRs.

The SFRs in (82) and (83) occur as the complement of existential 'have' and are interpreted as indefinites, as shown by the use of singular indefinites or bare plurals in the English translations as the heads of the headed RCs that are used to render the SFRs in Tigrinya.

- (82) [— dəq-ey z-i-kitatəl] **jε-bil-εji-n**
 children-POSS.1S REL-SM.3MS-look_after.IPFV NEG-have.PRES-SM.1S-NEG
 'I have no one to look after my children.' (Lit.: 'I don't have a person/people that look after my children.')
- (83) a. [— z-i-zarebi-lu] **jε-bil-εji-n**
 REL-SM.1S-talk.IPFV-OBL.3MS NEG-have.PRES-SM.1S-NEG
 'I have no one to talk to.' (Lit.: 'I don't have a person/people I talk to.')
- b. [— ni-wilad-εj k-i-zareb-o z-i-xiʔil]
 DAT-child-POSS.1S SBJ-SM.3MS-talk.IPFV-O.3MS REL-SM.3MS-can.IPFV
jε-bil-εji-n
 NEG-have.PRES-SM.1S-NEG
 'I have no one my child can talk to.'

One last example, in (84) shows an SFR as the complement of the intensional predicate 'want'. It is rendered in English with the RC headed by the indefinite *something*.

- (84) [— kibur z-εj-kon-ε] **ji-deli**
 expensive REL-NEG-become.PFV-SM.3MS SM.1S-**want**.IPFV
 ‘I want something that is not expensive.’¹²

This distributional and semantic pattern of SFRs is not surprising. The SFRs in the Mesoamerican languages in [Caponigro et al. 2021](#) exhibit the very same pattern. Also, free RCs across languages exhibit a similar pattern as well ([Caponigro 2003, 2004, 2021, 2023, Šimík 2011, 2017](#)). The emerging generalization seems to be that headless RCs without a D head semantically behave like referential nominals. They can only behave like quantificational nominals if they are introduced by a quantificational D head or occur as the complement of an existential predicate. There is another special case that escapes this generalization. We discuss it in the next section.

3.3 Free-choice Headless Relative Clauses

We have already established that the interpretation of LHRs with a demonstrative light head is always referential, similar to definite nominals in languages with a definite determiner. SFRs behave the same semantically, unless they occur as complements of existential predicates. There is another environment in which neither LHRs nor SFRs exhibit a semantic behavior that is plainly referential, like definite nominals — when they occur with the free choice marker *zixone*. Example (85), (86) and (87) show that the SFR and the LHR with *zixone* do not refer to a specific winner, but leave it open who the actual winner may be (‘free choice’). The variation in acceptability when the demonstrative light head is present is left to future research.

- (85) a. [(**zixone**) z-i-ɬawet] waga ji-rɛkib
 FREE_CHOICE REL-SM.3MS-win-IPFV price SM.3MS-get.IPFV
 b. %[?it-i (**zixone**) z-i-ɬawet] waga ji-rɛkib
 DIST-MS FREE_CHOICE REL-SM.3MS-win-IPFV price SM.3MS-get.IPFV
 ‘Whoever wins gets a price.’
- (86) a. [(**zixone**) ?ab kifil-εj zi-rɛkɛbi-w-o]
 FREE_CHOICE LOC room-POSS.1S REL-find.PFV-SM.1S-OM.3MS
 ji-nbib
 SM.1S-read.IPFV
 b. %[?it-i (**zixone**) ?ab kifil-εj zi-rɛkɛbi-w-o]
 DIST-MS FREE_CHOICE LOC room-POSS.1S REL-find.PFV-SM.1S-OM.3MS
 ji-nbib
 SM.1S-read.IPFV
 ‘I read whatever I find in my room.’
- (87) a. [nab (**zixone**) z-i-xɛjjid-o] ji-xɛjjid
 DIR FREE_CHOICE REL-SM.3MS-go.IPFV SM.1S-go.IPFV
 b. [nab-t-i (**zixone**) z-i-xɛjjid-o] ji-xɛjjid
 DIR-DIST-MS FREE_CHOICE REL-SM.3MS-go.IPFV SM.1S-go.IPFV
 ‘I go wherever he goes.’

Note that the LHR in (85b), (86b) and (87b), and the SFR in (85a), (86a) and (87a) without *zixone* allow for the very same ‘free choice’ interpretation as those with *zixone* — an interpretation that is not plainly referential. None of these headless RCs (with or without a demonstra-

¹²The root /kwn/ ‘become’ is the suppletive form of the verb ‘be’ used in negated and future contexts.

tive light head or *zixone*) refers to a specific individual that the speaker has in mind. Rather, they all convey some form of speaker's ignorance or indifference towards the specific individual whom those headless RCs are referring to, as long as it has the property of being the winner.¹³ Crucially, though, this 'free choice' reading is the only available reading for LHRs and SFRs with *zixone*, while those without easily allow for a referential interpretation, as shown earlier. Therefore, in a context that does not allow for 'free choice' variation in the reference of a headless RC, LHRs and SFRs with *zixone* are unacceptable, while those without *zixone* are. This is shown in (88) in which *nissu Tesfaj jibhal* 'his name is Tesfaj' does not allow for 'free choice variation': as a matter of fact, *zixone* in (88c) is ungrammatical; whereas, *niħade* 'one' in (88a) and *ni-t-i* in (88b) are perfectly grammatical.

- (88) a. [?it-i p'olis **ni-ħade** ni-ħanti məkina
DIST-MS police DOM-one.M DOM-one.F car.FS
zi-serek-Ø-a] ?asir-u-wo, nissu Tesfaj
REL-steal.PFV-SM.3MS-OM.3FS arrest.GER-SM.3MS-OM.3MS he Tesfaj
ji-bhal
SM.3MS-be_named.IPFV
'The police arrested one that stole one car; his name is Tesfaj.'
- b. [?it-i p'olis **ni-t-i** ni-ħanti məkina
DIST-MS police DOM-DIST-MS DOM-one.F car.FS
zi-serek-Ø-a] ?asir-u-wo, nissu Tesfaj
REL-steal.PFV-SM.3MS-OM.3FS arrest.GER-SM.3MS-OM.3MS he Tesfaj
ji-bhal
SM.3MS-be_named.IPFV
'The police arrested the one that stole one car; his name is Tesfaj.'
- c. *[?it-i p'olis **zixone** ni-ħanti məkina
DIST-MS police FREE_CHOICE DOM-one.F car.FS
zi-serek-Ø-a] ?asir-u-wo, nissu Tesfaj
REL-steal.PFV-SM.3MS-OM.3FS arrest.GER-SM.3MS-OM.3MS he Tesfaj
ji-bhal
SM.3MS-be_named.IPFV
('The police arrested one that stole one car; his name is Tesfaj.')

The free choice marker *zixone* is bimorphemic and consists of the relative marker *z(i)-* and the verbal root /kwn/, which functions in Tigrinya both as the lexical verb 'become' and as the suppletive form of the negative individual-level copula *?ijju*. The examples in (89)–(91) show that *zixone* can occur outside of headless RCs and realize different kinds of constituents: an inanimate direct object, the complement of a locative preposition, and the complement of a comitative preposition.

- (89) Dan **zixone** ji-belʃ
Dan FREE_CHOICE SM.3MS-eat.IPFV
'Dan eats anything.'
- (90) Dan ab **zixone** ji-deqis
Dan LOC FREE_CHOICE SM.3MS-sleep.IPFV
'Dan sleeps anywhere.'

¹³See Chierchia 2013 for a detailed proposal on how to model free choice as some form of existential quantification.

- (91) Dan mis **zixone** ji-zarib
 Dan with FREE_CHOICE SM.3MS-talk.IPFV
 ‘Dan talks with anyone.’

As a free-choice marker, *zixone* resists episodic and specific readings. This is illustrated by the contrast between the contexts provided for the two examples below. Example (92) yields a non-specific reading, whereas example (93) yields a specific one. Note that the specific reading in (93) requires the presence of a demonstrative beside *zixone*.

- (92) *Context: I am in the living room and cannot see my husband, who is in the kitchen. My husband really loves tomatoes. I think:*

zixone z-i-kiʃin ʔall-o, (naj giddin) komedere
 FREE_CHOICE SM.3MS-cook.IPFV BE₁.PRES-SM.3MS of obligation tomatoes
 all-ɛwu
 BE₁.PRES-SM.3MS

‘Whatever he is cooking, it must contain tomatoes.’ (Lit. ‘Whatever he is cooking, there are obligatorily tomatoes.’)

- (93) *Context: I am in the kitchen with my husband and my daughter. My husband is cooking, and I can see what he is cooking. My husband really loves tomatoes, and I can see that he is cooking some. I turn to my daughter and say:*

a. ***zixone** z-i-kiʃin ʔall-o, (naj giddin) komedere
 FREE_CHOICE SM.3MS-cook.IPFV BE₁.PRES-SM.3MS of obligation tomatoes
 all-ɛwu
 BE₁.PRES-SM.3MS

b. ʔit-i **zixone** z-i-kiʃin ʔall-o, (naj giddin)
 DIST-MS FREE_CHOICE SM.3MS-cook.IPFV BE₁.PRES-SM.3MS of obligation
 komedere all-ɛwu

tomatoes BE₁.PRES-SM.3MS

‘Whatever he is cooking, it must contain tomatoes.’ (Lit. ‘Whatever he is cooking, there are obligatorily tomatoes.’)

Note that from the presence of *zixone* in the grammatical sentence in (93b) that this element is therefore different from *anything* in English and free-choice free RCs in Italian and Romanian, which are infelicitous in contexts like (93) (Caponigro & Fălăuş 2018).

Zixone also cannot occur with quantifiers, as illustrated in (94).

- (94) *[bizuhat **zixone** z-i-ʃawet] waga ji-rəkib
 a_lot FREE_CHOICE REL-SM.3MS-win-IPFV price SM.3MS-get.IPFV
 Intended: ‘Many that win get a price.’

Lastly, *zixone* can appear in existential SFRs as well, as shown in (95).

- (95) [**zixone** z-i-zarebi-lu] jɛ-bil-ɛji-n
 FREE_CHOICE REL-SM.1S-talk.IPFV-OBL.3MS NEG-have.PRES-SM.1S-NEG
 ‘I have no one to talk to.’ (Lit.: ‘I don’t have anyone I talk to.’)

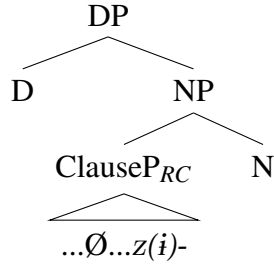
4 Conclusion

We have provided the first systematic description of headless RCs in Tigrinya, based on extensive novel fieldwork data. We have shown that Tigrinya employs a strategy that, to the best of

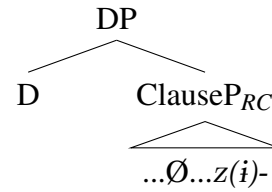
our knowledge, has not yet been described in the Afro-Asiatic literature. The language makes use of the typologically common strategy based on *wh*-words and *wh*-clauses to form matrix and embedded interrogative clauses. Headed RCs, instead, are formed by means of a different strategy, which does not make use of *wh*-words and uses a complementizer that is different from the one in interrogative clauses. This latter strategy, which is not based on *wh*-words nor *wh*-clauses, is the one adopted by all three varieties of headless RCs attested in Tigrinya: LHRs, SFRs, and free choice headless RCs. (The language has no FRs since they must be introduced by a *wh*-word by definition.)

Our findings can be schematized in a simplified way as in (96)–(99). Headed and headless RCs share the same morphosyntactic core: a non-*wh* clause (ClauseP_{RC}) characterized by the complementizer *z(i)-* and a missing argument. This structure is all that SFRs are made of, as shown in (97). LHRs add a D layer on top of it, as shown in (98), while headed RCs add both an NP and a DP layer, as shown in (96). By contrast, interrogative clauses are realized by means of a different clausal structure (ClauseP_{wh}) that is characterized by a different complementizer (*kɛmz(i)-*) and by overt *wh*-phrases in situ, rather than missing arguments, as shown in (99).

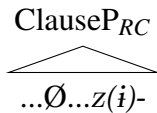
(96) *Headed RCs*



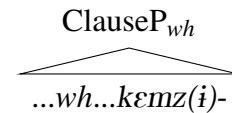
(98) *LHRs*



(97) *SFRs*



(99) *Wh-interrogatives*



Our findings about Tigrinya highlight a pattern that differs from those attested in other languages. Languages such as Abaza and Adyghe do not use *wh*-clauses nor *wh*-words to form any kind of RC (including SFRs), resembling Tigrinya. On the other hand, they do not use the *wh*-strategy for (embedded) interrogative clauses either, unlike Tigrinya. At the opposite side of the spectrum, languages like Polish and Romanian make extensive use of *wh*-clauses and *wh*-words to form interrogative and non-interrogative clauses, but they lack SFRs. In the middle, a language like Maltese, which has headless RCs with *wh*-elements (FRs) and without *wh*-elements (SFRs).

By documenting the Tigrinya pattern, our study refines how relative clause formation is typologically derived and highlights a previously understudied strategy, which is described in Afro-Asiatic for the first time (to the best of our knowledge).

The descriptive generalizations we have established provide the empirical groundwork for future research focused on a detailed, formal syntactic and semantic analysis of these constructions. We also hope to have provided a model for future descriptive work on headless RCs in other Ethio-Semitic and Afro-Asiatic languages — work that would contribute to strengthening and broaden our current understanding and typology of headless RCs and RCs in general.

References

- Ahn, Dorothy. 2022. Indirectly direct: An account of demonstratives and pointing. *Linguistics and Philosophy* 1345–1393.
- Appleyard, David. 2015. Ethiopian semitic and cushitic: An ancient contact features in ge'ez and amharic. In *Semitic Languages in Contact*, ed. Aaron Michael Butts, Studies in Semitic Languages and Linguistics, 16–32. Leiden/Boston: BRILL.
- Arkadiev, Peter, & Ivano Caponigro. 2021. Conveying content questions without wh-words: evidence from abaza. In *Proceedings of Sinn und Bedeutung*, ed. Patrick Grosz, Luisa Marti, Hazel Pearson, Yasutada Sudo, & Sarah Zobel, volume 25, 73–94. URL <https://doi.org/10.18148/sub/2021.v25i0.925>.
- Bossong, Georg. 1991. Differential object marking in romance and beyond. In *New Analyses in Romand Linguistics*, ed. Wanner Dieter & Douglas A. Kibbee, 143–170. Amsterdam: John Benjamins Publishing Company.
- Bulakh, Maria. 2019. Tigrinya. In *The Semitic Languages*, ed. John Huehnergard & Na'ama Pat-El, 174–201. London New York: Routledge.
- Cacchioli, Gioia. 2023a. Subordination as relativization: The case of tigrinya.
- Cacchioli, Gioia. 2023b. The tigrinya *zi-* prefix: A morphological reflex of successive-cyclic movement. *Brill's Journal of Afroasiatic Languages and Linguistics* 15:232–261.
- Cacchioli, Gioia. 2026. The syntax of clausal prefixes in tigrinya. Ph.d. thesis, Université de Genève, Genève.
- Cacchioli, Gioia, & Jason D. Overfelt. under review. Topics in the syntax of tigrinya. In *Handbook of Afroasiatic Languages and Linguistics*, ed. Karim Bensoukas & Alireza Korangy, Springer Handbooks of Languages and Linguistics. Springer.
- Caponigro, Ivano. 2003. Free not to ask: On the semantics of free relatives and wh-words cross-linguistically. Ph.d. thesis, University of California, Los Angeles.
- Caponigro, Ivano. 2004. The semantic contributions of wh-words and type shifts: Evidence from free relatives crosslinguistically. *Semantics and Linguistic Theory* 14:38–55.
- Caponigro, Ivano. 2021. Introducing headless relative clauses and the findings from mesoamerican languages. In *Headless relative clauses in Mesoamerican languages*, ed. Ivano Caponigro, Harold Torrence, & Roberto Zavala Maldonado, 1–57. Oxford: Oxford University Press.
- Caponigro, Ivano. 2023. Referring and quantifying without nominals: Headless relative clauses across languages. In *Proceedings of the 33rd Semantics and Linguistic Theory Conference (SALT 33)*, ed. Juhyae Kim, Burak Öney, & Fengyue (Lisa) Zhao, 744–774.
- Caponigro, Ivano, & Anamaria Fălăuș. 2018. Free choice free relative clauses in italian and romanian. *Natural Language Linguistic Theory* 36:323–363.
- Caponigro, Ivano, & Anamaria Fălăuș. 2023. Investigating clausal wh-constructions in romanian. In *Romance languages: recent contributions to linguistic theory*, ed. Roberta Pires de Oliveira, volume 9, 1–32. Isogloss. Open Journal of Romance Linguistics. Special Issue.

- Caponigro, Ivano, & Maria Polinsky. 2011. Relative embeddings: a circassian puzzle for the syntax/semantics interface. *Natural Language Linguistic Theory* 29:71–122.
- Caponigro, Ivano, Harold Torrence, & Roberto Zavala Maldonado. 2021. *Headless relative clauses in mesoamerican languages*. Oxford: Oxford University Press.
- Chierchia, Gennaro. 2013. *Logic in grammar: Polarity, free choice, and intervention*. Oxford: Oxford University Press.
- Citko, Barbara. 2004. On headed, headless, and light-headed relatives. *Natural Language Linguistic Theory* 22:95–126.
- Gebregziabher, Keffyalew, & Maia Duguine. 2024. Typology of tigrinya wh-interrogatives.
- Kievit, Dirk, & Saliem Kievit. 2009. Differential object marking in tigrinya. *Journal of African Language and Linguistics* 30:45–71.
- King, Jeffrey C. 2001. *Complex demonstratives: A quantificational account*, volume 2. Cambridge (Mass.): MIT Press.
- Kogan, Leonid E. 1997. Tigrinya. In *The Semitic Languages*, ed. Robert Hetzron, 424–445. London: Routledge.
- Leslau, Wolf. 1941. *Documents tigrinya (Éthiopien septentrional): Grammaire et textes*. Paris: Librairie C. Klincksieck.
- Mantenuto, Iara, & Ivano Caponigro. 2021. Light-headed relative clauses in teramano. *Studies in Language* 45:921–937.
- Mason, John. 1996. *Tigrinya grammar*. Laurenceville: The Red Sea Press, 2 edition. First published 1994 by the publisher.
- Meyer, Ronny, & Yvonne Treis. 2021. How to quote ethiopian authors in linguistic publication. *Afrika und Übersee* 94:80–90.
- Nazareth Amlesom Kifle. 2011. Tigrinya applicatives in lexical-functional grammar. Ph.d. thesis, University of Bergen.
- Overfelt, Jason D. 2009. The syntax of relative clause constructions in tigrinya. Ma thesis, Purdue University.
- Overfelt, Jason D. 2022. Asymmetrical symmetry in tigrinya object marking. In *Angles of Object Agreement*, ed. Andrew Nevins, Anita Peti-Stantic, & Mark de Vos, 135–163. Oxford: Oxford University Press.
- Palmer, Frank R. 1955. The ‘broken plurals’ of tigrinya. *Bulletin of the School of Oriental and African Studies* 17:548–566.
- Palmer, Frank R. 1962. Relative clauses in tigrinya. *Journal of Semitic Studies* 7:36–43.
- Sadler, Louisa, & Maris Camilleri. 2018. Free relatives in maltese. *Brill’s Journal of Afroasiatic Languages and Linguistics* 10:115–159.
- Simonenko, Alexandra. 2014. Grammatical ingredients of definiteness. Ph.d. thesis, McGill, Montréal (Canada).

- Tesfay Tewolde Yohannes. 2016. Dps, phi-features and tense in the context of abyssinian (eritrean and ethiopian) semitic languages: A window for further research. Ph.d. thesis, Università di Firenze, Firenze.
- van Urk, Coppe. 2024. The morphological consequences of complementizer placement in tigrinya. In *Talk at Affixation in Afroasiatic*. Paris.
- van Urk, Coppe. 2025. Intermediate steps of lowering in tigrinya. URL <https://ling.auf.net/lingbuzz/009378>, London.
- de Vries, Mark. 2002. The syntax of relativization. Ph.d. thesis, Amsterdam University, Amsterdam.
- Weldeyesus, Weldu M. 2004. Case marking systems in two ethiopian semitic languages. *Colorado Research in Linguistics* 17.
- Weninger, Stefan. 2011. Ethio-semitic in general. In *The Semitic Languages: An International Handbook*, ed. Stefan Weninger, 1114–1122. Berlin, Boston: De Gruyter Mouton.
- Šimík, Radek. 2011. Modal existential wh-constructions. Ph.d. thesis, University of Groningen, Holland.
- Šimík, Radek. 2017. Existential wh-constructions. In *Oxford bibliographies in linguistics*, ed. Mark Aronoff. New York: Oxford University Press.